

A “Quarto” Life Crisis

Are you still living the “Copy-Paste” lifestyle? Moving data into tables, images and screenshots into folders, and repeating each time changes are applied?

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Quarto

- enables you to weave together content and executable code into a finished presentation.
- makes the process of creating and collaborating on scientific and technical documents dramatically better.
- is Posit's attempt to bring R Markdown to everyone! Unlike R Markdown, Quarto doesn't have a dependency or requirement for R.

Minimum requirements

for writing research papers



Figure 1: R Language



Figure 2: RStudio IDE



Figure 3: TinyTeX

i Note

RStudio v2022.07 and later includes support for editing and preview of Quarto documents.

Integrated workflow

R Markdown and Quarto allow for the seamless integration of narrative text, raw data, and executable code in a single document.

Simple Markdown syntax

Table 1: LaTeX vs Markdown

Feature	LaTeX	Quarto
Bold	<code>\textbf{Heavy}</code>	<code>**Heavy**</code>
<i>Italic</i>	<code>\textit{Slanted}</code>	<code>*Slanted*</code>

Feature	LaTeX	Quarto
Links	<code>\href{url}{text}</code>	<code>[text](url)</code>
Code	<code>\texttt{fixed}</code>	<code>`fixed`</code>
Images	<code>\includegraphics{...}</code>	<code></code>

Verdict: Less typing, more presenting for Quarto.

Inline and Display Math

- In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

- $$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS \frac{\partial C}{\partial S} = rC \quad (1)$$

- $$\begin{aligned} f(x) &= a(x - h)^2 + k \\ &= a(x^2 - 2xh + h^2) + k \end{aligned} \quad (2)$$

Code execution control

- **echo: Show the source code.** If false, the code stays hidden.

```
– #| echo: true
```

```
print("I want to show the codes.")
```

```
[1] "I want to show the codes."
```

```
– #| echo: false
```

```
[1] "I want only the result."
```

- **eval: Evaluate the code.** If false, the code is never executed.

```
– #| eval: true
```

```
print("Run the code.")
```

```
[1] "Run the code."
```

```
– #| eval: false
```

```
print("Don't run the code.")
```

Dynamic updating

Results, tables, and figures update automatically when the underlying data changes, eliminating manual editing errors and ensuring the final paper remains up to date with the analysis.

Input

```
1 # load data
2 library("kableExtra")
3 data(iris)
4 iris2 <- iris[1:3, c(1, 3, 2, 4, 5)]
5
6 # calculations
7 mean_sepal_length <- mean(iris2$Sepal.Length)
8 mean_sepal_width <- mean(iris2$Sepal.Width)
9
10 # format into a pretty table
11 names(iris2) <- gsub('[.].+', '', names(iris2))
12 rownames(iris2) <- NULL
13 iris_table <- kable(iris2, booktabs = TRUE) %>%
14   add_header_above(c("Length" = 2, "Width" = 2, " " = 1)) %>%
15   add_header_above(c("Measurements" = 4, "More attributes" = 1))
```

Output

The means of sepal length and sepal width are 4.9 and 3.23, respectively.

Table 2: Iris Data

Measurements				More attributes
Length		Width		
Sepal	Petal	Sepal	Petal	Species
5.1	1.4	3.5	0.2	setosa
4.9	1.4	3.0	0.2	setosa

Multi-format outputs

One source file can be rendered into professional PDFs, Word documents or Tex files for collaborators, interactive HTML reports and presentations, as well as into beamer or PowerPoint.

Flexibility

1. Specify the formats in your YAML.
2. Use the **Render** button or the terminal (`quarto render your_qmd.qmd --to format`).
 - try for beamer
 - try for html
 - try for pdf

Automated elements

Complex tasks such as cross-referencing (See Section , see [Cross-referencing](#)), bibliography management (See Section), journal formats (See Section), plots and table formatting are handled programmatically, which maintains consistency throughout long documents.

Cross-referencing

Cross-reference	Text
Equations	Black-Scholes (Equation 1) is a mathematical model... Equation 2 shows...
Image	See Figure 2
Tables	Table 1 shows how LaTeX differs from Markdown. The data is shown in Table 2

Citations

- Bibliography file (`.bib`)
- Syntax: According to @klein -> According to Klein, Wright, and Wieczorek (2020)
- YAML
 - `link-citations: true`
 - `nocite` key in YAML
- Citation style (`.csl`)

Extensions for journals

JASA

Challenges

- Debugging
 - error messages can be vague (especially YAML + LaTeX)
 - one change may silently override defaults
- Formatting differences across outputs
 - looks great in RevealJS but breaks in PDF
 - fonts, spacing, figure placement behave differently
- File organization
 - managing images, references, and extensions can get messy

References

- jpatteet. 2022. “Quarto Rendering Pdf, Running Xelatex 10 Times.” November 22, 2022. <https://github.com/quarto-dev/quarto-cli/issues/3518>.
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- Posit. n.d. “Comprehensive Guide to Using Quarto.” <https://quarto.org/docs/guide/>.